COEQWAL RESULTS BRIEF · STRATEGY

Prioritizing full demands of community water systems: modeled outcome levels across five hydroclimates, and how much of the difference from the baseline is the hydroclimate

A results brief for a general reader — prototype (v0_3), built by the brief builder from the locked 2026-08-29 API snapshot — reporting modeled outcome levels for one management strategy read across the five hydroclimates, each difference from the baseline split into the two effects that produce it. The build stamp, the run identifiers and the verification status are in references.

TL;DR. Compared with the baseline — current operations under the historical hydroclimate — this strategy's modeled outcome levels differ from the baseline in all nine sectors of the water system under at least one hydroclimate. Reading each sector under each hydroclimate as one result, 45 in all, the hydroclimate effect is the larger of the two in 35, and the operations effect in 10. The strategy's value differs from current operations at 20 of the 279 locations reported here under the historical hydroclimate and at 29 under extreme climate stress, and across the five hydroclimates the count ranges from 20 to 41; where it does not differ, the difference from the baseline is the hydroclimate alone. Community surface water is the clearest case of the two effects running opposite ways: under high climate stress the hydroclimate contributes +0.5 and the strategy −0.2, leaving a difference from the baseline of +0.3.

What this is

This is a *results brief*: it reports the modeled outcome levels, and nothing else, for the prioritizing full demands of community water systems strategy — one of 29 management strategies in the COEQWAL scenario library — simulated once under each of five hydroclimates and read as a set. The selection rule is that there is none: this brief reports every sector of the water system these model runs carry an outcome level for, across all five modeled hydroclimates, and no filter is applied to either. All nine sectors of the water system are reported under every hydroclimate. A scenario is a strategy paired with one hydroclimate; the five hydroclimates are a sample of stress levels, not a forecast of which will occur. For how the five were constructed, see the hydroclimates reference brief.

Against current operations, this strategy differs in one of the eight operational settings the scenario library records: Allocation / priorities — Top priority for full contract demand level of surface deliveries to CVP and SWP M&I contractors. For the full configuration, see the prioritizing full demands of community water systems strategy brief.

What the model shows

Across the nine sectors these runs report, the modeled level values under the baseline — current operations under the historical hydroclimate — range from 1.7 to 4.2. Read across the five hydroclimates, the values this brief reports range from 1.6 to 4.8. Of the 45 results — one sector under one hydroclimate — 15 sit in a different level band from that sector's baseline value and 30 sit in the same band; in eight of the nine sectors the five values do not all sit in one band.

The figure below draws every one of those results beside the baseline, with the two effects that produce the difference shown as separate marks, so a reader can see how much of each difference the hydroclimate carries and how much the strategy does. The section that follows says what the two effects are before the figure draws them; what the level values themselves mean is set out in how to read the values.

The baseline, and the two effects

The baseline is current operations under the historical hydroclimate. It is the one reference this brief asks a reader to learn, and it is the same value in every column, because it does not depend on which hydroclimate a scenario pairs with.

Any difference between a scenario and the baseline comes from two places, and they add exactly:

The hydroclimate effect is current operations under that hydroclimate minus the baseline — the same operations, a different hydroclimate. **The operations effect** is this strategy minus current operations *under that same hydroclimate* — the only comparison that holds the hydroclimate fixed, and therefore the only one that isolates what the strategy does. Together they equal the difference from the baseline.

Outcome levels are reported as a value on a continuous axis from Level 1 to Level 4.99, with the four level bands as shaded zones; the whole number below any value is its band. A value that lands exactly on a whole number takes that band: 4.0 is Level 4, at the start of its zone rather than the top of Level 3. The level scale is ordered by definition, Level 1 through Level 4; a band is a category and a position on the axis is a modeled magnitude, neither is a judgment. The scale, the two thresholds and the two fields a level value comes in are set out in how to read the values.

A sector's level value is the mean of the level values at its locations. The nine sectors differ in how many locations that is. Delta estuary ecology is reported at one location, Delta-Eastside Water. Winter-run salmon is reported at one location, Sacramento River at Keswick. Freshwater for Delta exports is reported at two locations, Harvey O. Banks Pumping Plant and C.W. "Bill" Jones Pumping Plant (formerly Tracy). Both carry the same level value in every scenario in this ensemble. Freshwater for in-Delta uses is reported at two locations, Emmaton and

Jersey Point. Both carry the same level value in every scenario in this ensemble. The remaining five sectors are reported across 8 to 132 locations. On the figure, a square mark identifies the sectors reported at one or two locations. This page reports sectors, not individual locations; where a place brief has been published for one of a sector's locations, that brief reports the location's own values across the same scenarios.

All nine sectors, all five hydroclimates

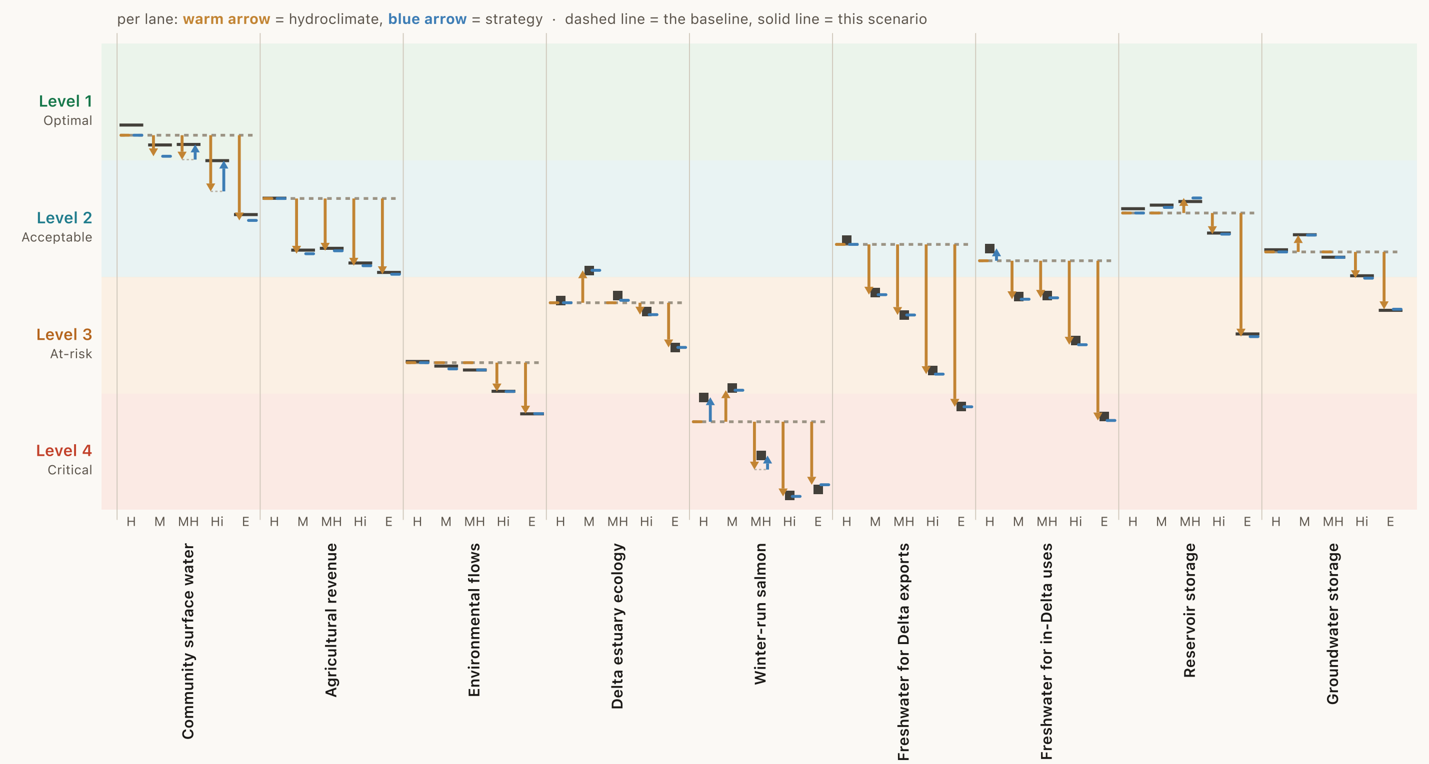

Figure 1. Modeled level values for prioritizing full demands of community water systems and the two effects that produce their difference from the baseline, nine sectors of the water system × five hydroclimates. Lanes within each sector run left to right: Historical (H), Moderate (M), Moderate-high (MH), High (Hi), Extreme (E) climate stress. Vertical axis: continuous level value, Level 1 at top, level bands as shaded zones. Dashed level: the baseline. Solid level: this scenario. A short flat tick in place of an arrow is a change smaller than 0.1 on the level scale. Square marks are the sectors reported at one or two locations. Every value is in the table that follows, and the scale is set out in how to read the values.

The same 45 results as numbers

Each cell reads top to bottom: the hydroclimate effect, the operations effect, and the resulting value with its band. The two effects add to the difference between the scenario and the baseline. Outlined cells are those where the two run opposite ways. Each hydroclimate column carries its own address for citation, and its own slice of the companion data file.

Sector	The baseline	Historical	Moderate	Moderate-high	High	Extreme	Operations effect across the five
Community surface water	L1 1.7	0.0 0.0 L1 1.6	+0.2 0.0 L1 1.8	+0.2 −0.1 L1 1.8	+0.5 −0.2 L2 2.0	+0.8 0.0 L2 2.4	one direction −0.2 to 0.0
Agricultural revenue	L2 2.3	0.0 0.0 L2 2.3	+0.5 0.0 L2 2.7	+0.4 0.0 L2 2.7	+0.6 0.0 L2 2.8	+0.6 0.0 L2 2.9	no change at the 0.1 threshold 0.0
Environmental flows	L3 3.7	0.0 0.0 L3 3.7	0.0 0.0 L3 3.7	0.0 0.0 L3 3.7	+0.2 0.0 L3 3.9	+0.4 0.0 L4 4.1	no change at the 0.1 threshold 0.0
Delta estuary ecology SINGLE LEVEL VALUE	L3 3.2	0.0 0.0 L3 3.2	−0.3 0.0 L2 2.9	0.0 0.0 L3 3.1	+0.1 0.0 L3 3.2	+0.4 0.0 L3 3.6	no change at the 0.1 threshold 0.0
Winter-run salmon SINGLE LEVEL VALUE	L4 4.2	0.0 −0.2 L4 4.0	−0.3 0.0 L3 3.9	+0.4 −0.1 L4 4.5	+0.6 0.0 L4 4.8	+0.5 0.0 L4 4.8	one direction −0.2 to 0.0
Freshwater for Delta exports SINGLE LEVEL VALUE	L2 2.7	0.0 0.0 L2 2.6	+0.4 0.0 L3 3.1	+0.6 0.0 L3 3.3	+1.1 0.0 L3 3.8	+1.4 0.0 L4 4.1	no change at the 0.1 threshold 0.0
Freshwater for in-Delta uses SINGLE LEVEL VALUE	L2 2.8	0.0 −0.1 L2 2.7	+0.3 0.0 L3 3.1	+0.3 0.0 L3 3.1	+0.7 0.0 L3 3.5	+1.4 0.0 L4 4.1	one direction −0.1 to 0.0
Reservoir storage	L2 2.4	0.0 0.0 L2 2.4	0.0 0.0 L2 2.3	−0.1 0.0 L2 2.3	+0.2 0.0 L2 2.6	+1.1 0.0 L3 3.4	no change at the 0.1 threshold 0.0
Groundwater storage	L2 2.7	0.0 0.0 L2 2.7	−0.1 0.0 L2 2.6	0.0 0.0 L2 2.8	+0.3 0.0 L2 2.9	+0.5 0.0 L3 3.2	no change at the 0.1 threshold 0.0

How many places a strategy reaches

A strategy's rules apply in some places and not others, and a sector's value is the unweighted mean of its locations. This strategy's value differs from current operations by at least 0.1 on the level scale at 14–31 of the 74 Community surface water locations, 0–12 of the 132 Agricultural revenue locations, 0–1 of the 17 Environmental flows locations, 0 of the one Delta estuary ecology location, the one Winter-run salmon location under some hydroclimates and not others, 0 of the 2 Freshwater for Delta exports locations, as few as 0 and as many as all 2 Freshwater for in-Delta uses locations, 1–2 of the 8 Reservoir storage locations and 0–2 of the 42 Groundwater storage locations, depending on the hydroclimate — 20 to 41 of the 279 locations reported here. A sector where few locations differ therefore shows a small sector-level difference even when the locations that do differ move by a full level or more. Read across the five hydroclimates together, 144 of the 1,395 location-and-hydroclimate results reported here differ from current operations: 120 toward Level 1, 24 toward Level 4. The median difference over the locations counted runs from 0.1 on the level scale at Freshwater for in-Delta uses to 0.3 at Community surface water. The locations that differ do not all differ in the same direction: in 8 of the 45 sector-and-hydroclimate combinations reported here, some of a sector's locations differ from current operations toward Level 1 and others toward Level 4, so a sector value can sit close to the baseline while its locations differ in both directions. This page reports sectors, not individual locations; where a place brief has been published for one of a sector's locations, that brief reports the location's own values across the same scenarios.

Related briefs

What does this strategy do? It is described, rather than measured, by a reference brief that carries no modeled results: Prioritizing full demands of community water systems strategy brief.

What happens under a different strategy? The same nine sectors and the same five hydroclimates are reported once per strategy, and once with the strategy held fixed:

- Current operations
- Current operations with historical land use
- Current operations with reintroduction
- Current operations without TUCPs
- Current USBR operations
- Current USBR operations without TUCPs
- Delta Conveyance Project
- Functional environmental flows
- Functional environmental flows with reduced crop acreage
- Functional environmental flows with reduced crop acreage and reintroduction
- Functional environmental flows with reintroduction
- Groundwater pumping limits in the Central Valley
- Groundwater pumping limits in the San Joaquin Valley
- Groundwater pumping limits via reduced crop acreage in the Central Valley
- Groundwater pumping limits via reduced crop acreage in the San Joaquin Valley
- Increase Delta outflows (55% unimpaired flow target)
- Increase Delta outflows (65% unimpaired flow target)
- Increase Shasta carry-over storage
- Maintain Delta outflows (45% unimpaired flow target)
- No flow requirements
- Prioritizing full demands of community water systems — *you are here*
- Prioritizing human health water deliveries to community water systems
- Reduce Delta outflows (35% unimpaired flow target)
- Relax Delta salinity standards
- The water system under current operations
- Winter-run refuge flows

Winter-run refuge flows with reduced crop acreage

Winter-run refuge flows with reduced crop acreage and reintroduction

Winter-run refuge flows with reintroduction

What happens to one sector under every strategy? A sector brief transposes this grid — one sector of the water system, every strategy, the same five hydroclimates:

Agricultural revenue

Community surface water

Delta estuary ecology

Environmental flows

Freshwater for Delta exports

Freshwater for in-Delta uses

Groundwater storage

Reservoir storage

Winter-run salmon

How were the five hydroclimates constructed? The hydroclimates reference brief defines them. It reports no results of its own; every brief above is read against the same five.

What this doesn't say

> This brief reports modeled results from the COEQWAL ensemble. It does not predict what will happen, evaluate whether those results are acceptable, or recommend any course of action. The outcome-level framework provides definitional categories; the meaning of those categories for any individual reader, community, or decision is not specified by COEQWAL.
>
> In particular, for this brief: the five hydroclimates are a sample of stress levels, carry no probabilities, and are not a timeline, so no average over them is reported here. Splitting a difference into two effects is arithmetic on modeled values, not a causal claim about the water system, and the two are separable only because each is defined against its own reference. A step or a spread on the level axis is a modeled magnitude, not a benefit or a cost. A sector's value is the unweighted mean of its locations, so locations count equally regardless of the people, acres, or acre-feet each represents, and a location already at either end of the scale cannot move further along it. Locations are modeled nodes, not measured conditions at a place.
>
> Extreme climate stress holds the single highest of a sector's five values in eight of the nine sectors reported here. In one the highest falls under a different hydroclimate. So where a sector's highest value falls is not settled by construction.

References, run identifiers and the build stamp

Model runs (this strategy / current operations): Historical `s0037` / `s0020`; Moderate `s0149` / `s0134`; Moderate-high `s0077` / `s0047`; High `s0097` / `s0056`; Extreme `s0123` / `s0108`. The baseline is current operations under the historical hydroclimate, `s0020`.

Build stamp. Snapshot `2026-08-29`; builder version `0.1.0`; built 2026-08-29T16:19:04Z; snapshot data read 2026-08-29; verification: pass. The verification status is written by the independent check (`briefbuilder.verifier`), which recomputes every value in the companion data file from the per-location data and reads every arrow on this page, after both files are written. It is not stamped by whatever produced them.

Companion data file: `results_prioritizing-full-demands-of-community-water-systems_schema1.json`, holding every value in this brief plus the per-location counts, the spread measures, the derivation of every number in the prose and the figure's declared attribute contract. Per-hydroclimate slices: Historical `#/rows/*/by_hydroclimate/BL`; Moderate `#/rows/*/by_hydroclimate/HC4`; Moderate-high `#/rows/*/by_hydroclimate/HC1`; High `#/rows/*/by_hydroclimate/HC2`; Extreme `#/rows/*/by_hydroclimate/HC3`. Built from the locked 2026-08-29 API snapshot (api.coeqwal.org v2.1.1 (locked dataset snapshot)). Decomposition per `Decomposition_Math_Quickref` (coeqwal-viz): the hydroclimate effect is current operations under that hydroclimate minus current operations under the historical hydroclimate; the operations effect is this strategy minus current operations under that same hydroclimate. Axis convention [1, 4.99] per `Ordinal_vs_continuous_tiers.md`; the visible band is `floor(value)`. A difference of 0.1 on the level scale is required before a difference is reported as movement, before a location is counted as differing from its comparison, and before an effect prints as anything other than 0.0. Color semantics (warm = hydroclimate, blue = operations) follow the ops-vs-climate stepper. Strategy identity is its descriptive name; run IDs appear only in this note and the data file.

Explore further

Every value on this page comes from the outcome-level data, and the data explorer holds the same data without this brief's selection of it. The link below opens prioritizing full demands of community water systems across all five hydroclimates and all nine sectors, so a reader can take the comparison apart, add a hydroclimate or a sector this brief reports as one number, and read the locations behind a sector value directly.

Open prioritizing full demands of community water systems in the data explorer.

How to read the values

The level scale runs from 1 to 4.99. Level 1 is the favorable end of the scale: the outcome-level framework names Level 1 Optimal and Level 4 Critical, so a smaller value sits nearer Optimal and a larger value nearer Critical. The API serves this number in a field named weighted_score; that name describes how the number is computed, and does not mean that a larger number is a more favorable result.

Two numbers describe each result. The integer tier_level is the level label. tier_continuous is the position within the band of that label, and always falls inside it: tier_continuous is at least tier_level and less than tier_level plus one. Labels, counts and prose use tier_level; figure positions and statistics use tier_continuous.

Level 1 is Optimal, Level 2 is Acceptable, Level 3 is At-risk, and Level 4 is Critical. These are definitional category names supplied by the outcome-level framework; what a category means for any reader, community, or decision is not specified by COEQWAL.

A sector value is the unweighted mean over the locations in that sector. Every location counts equally, regardless of the people, acres, or acre-feet it represents.

Any difference between two values under different hydroclimates contains both a hydroclimate effect and an operations effect; the two are separable only through the decomposition, because each is defined against its own reference. A strategy is compared with current operations under the same hydroclimate. Comparing one strategy with itself across hydroclimates is a different quantity: the hydroclimate effect plus the change in the operations effect.

One threshold governs every claim and every count here. A difference smaller than 0.1 on the level scale is not reported as movement, and a location is not counted as differing from its comparison. An outcome level is a summary built so that results in nine sectors with no shared unit can be compared, and 0.1 on the level scale is the finest reading it carries; a smaller difference is left unstated rather than described.

The five hydroclimates are a purposive sample of climate stress levels. They are not a sequence, a trend, or a forecast, they carry no probabilities, and no mean is taken over them.

A run ID identifies one modeling run of one strategy under one hydroclimate. It is an internal identifier, used for citation and for tracing a value back to the data platform, not a name for the strategy; the public identity of a strategy is its descriptive name.

Level values are reported to one decimal place throughout this brief — in the prose, in the figure and in the table. The level scale does not carry meaning finer than that. Each effect shown is the difference between the floored values, except that an effect smaller than 0.1 on the level scale, which is the threshold this brief reports movement at, is shown as 0.0, so a row containing one does not add up. Values are floored rather than rounded, so a printed value stays inside the band its own label names; full-precision values are in the machine-readable sidecar this brief is built from.